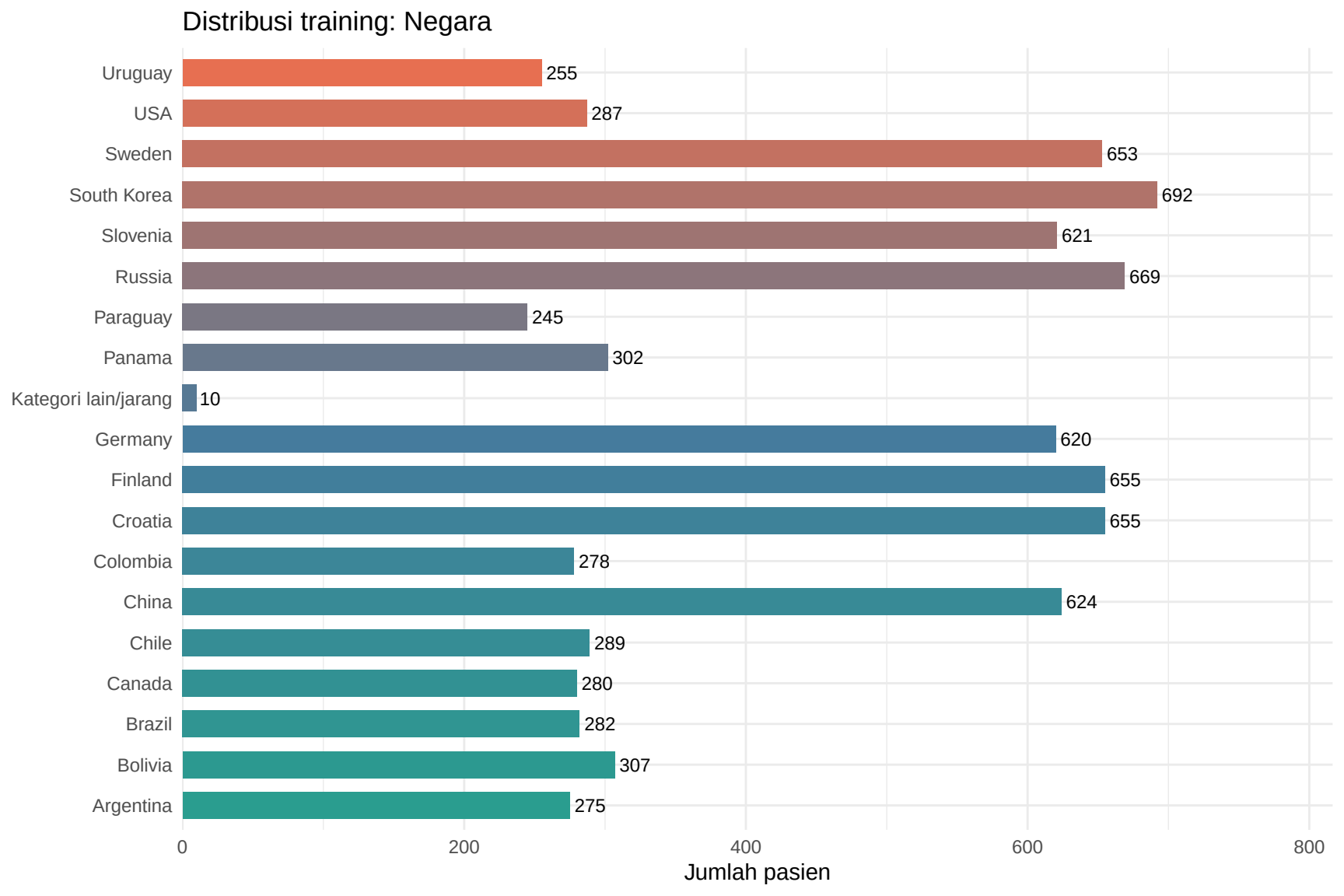
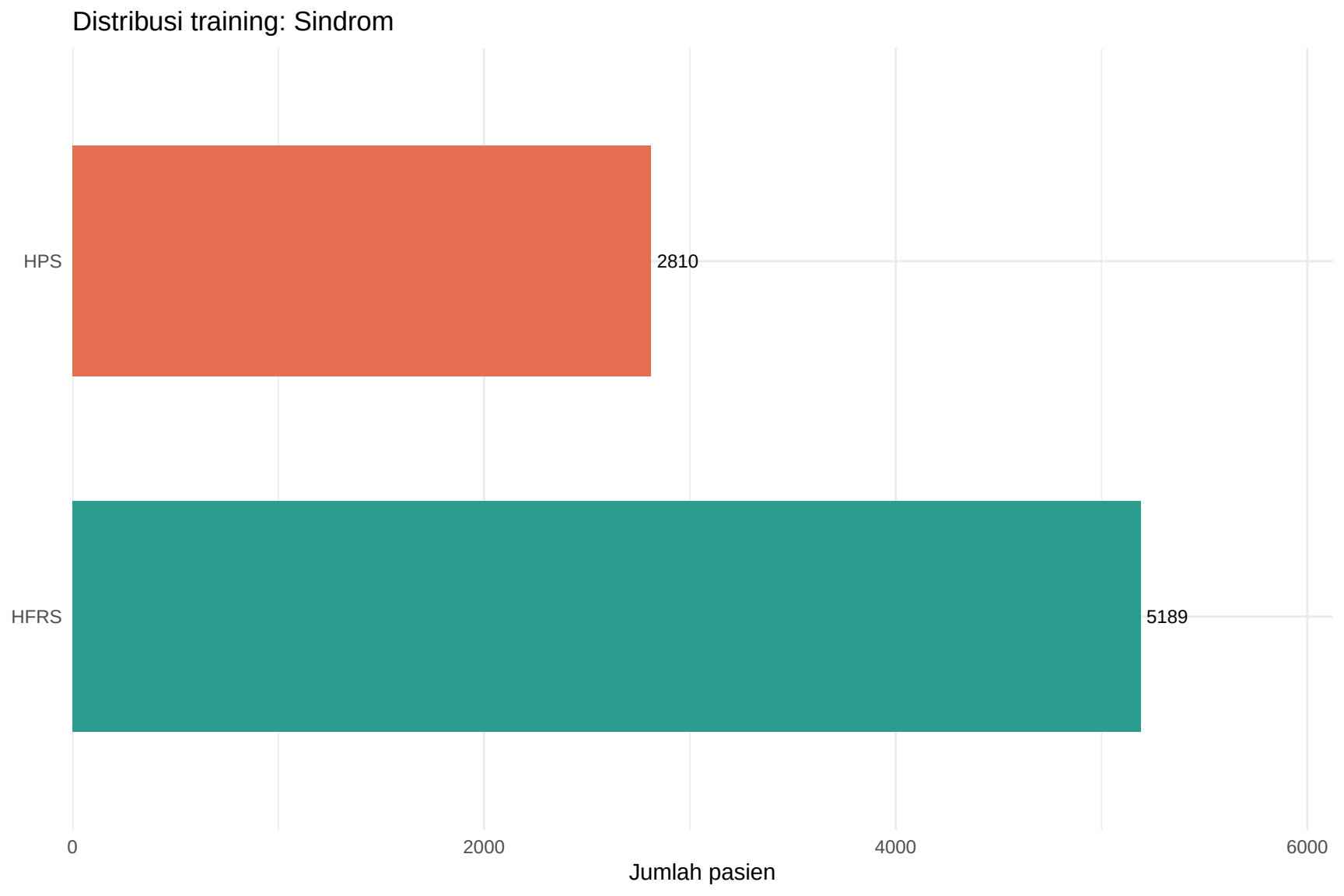
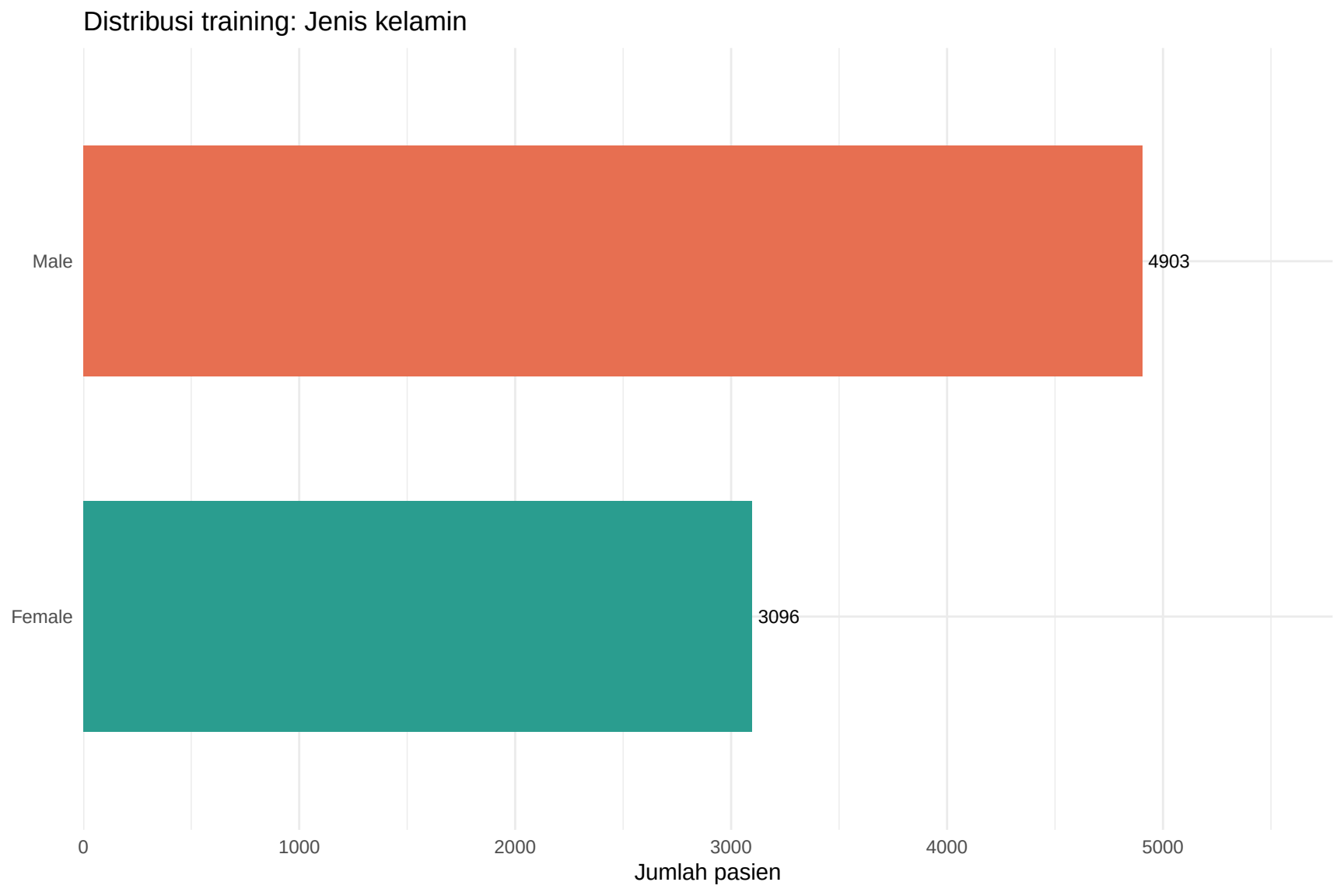
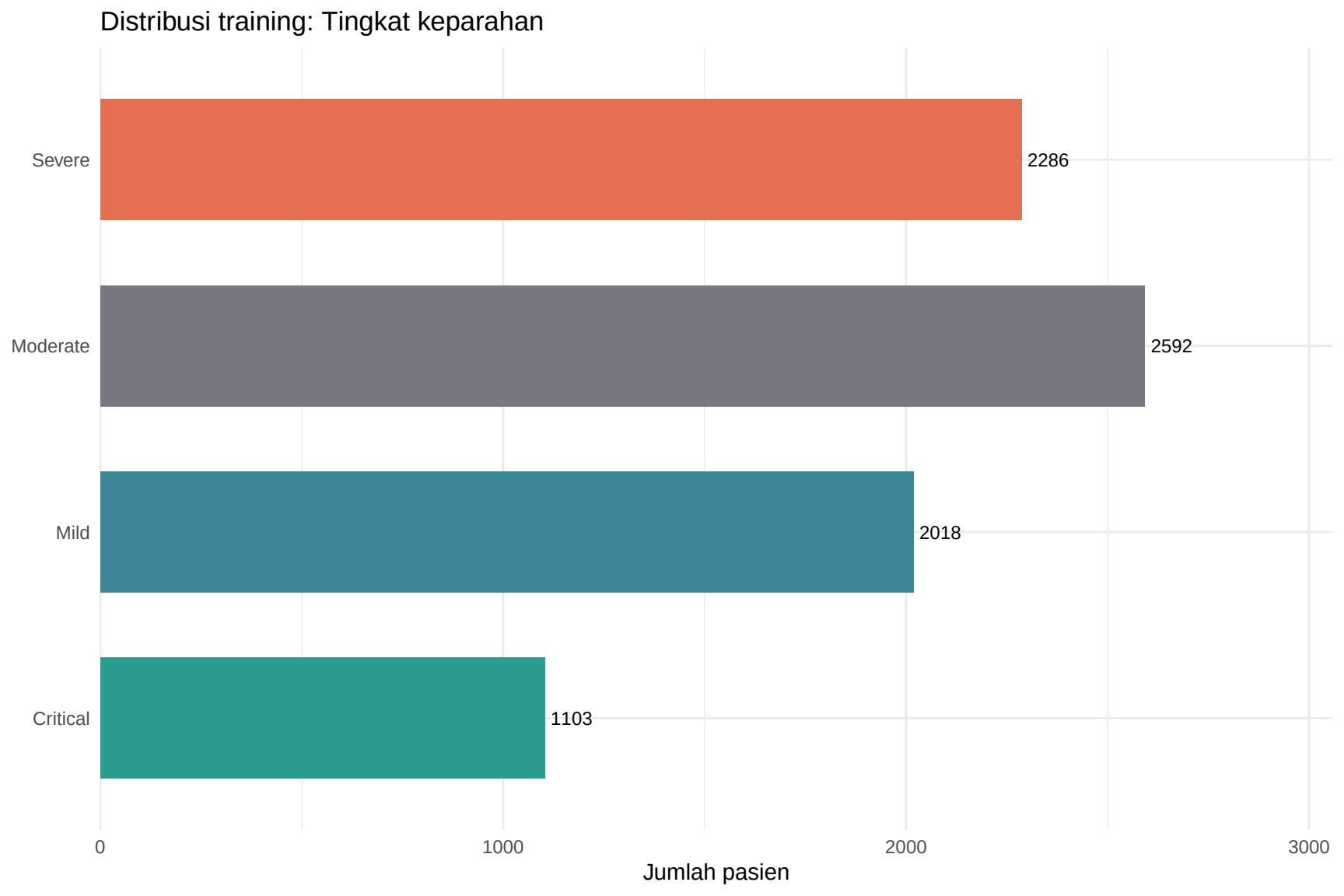
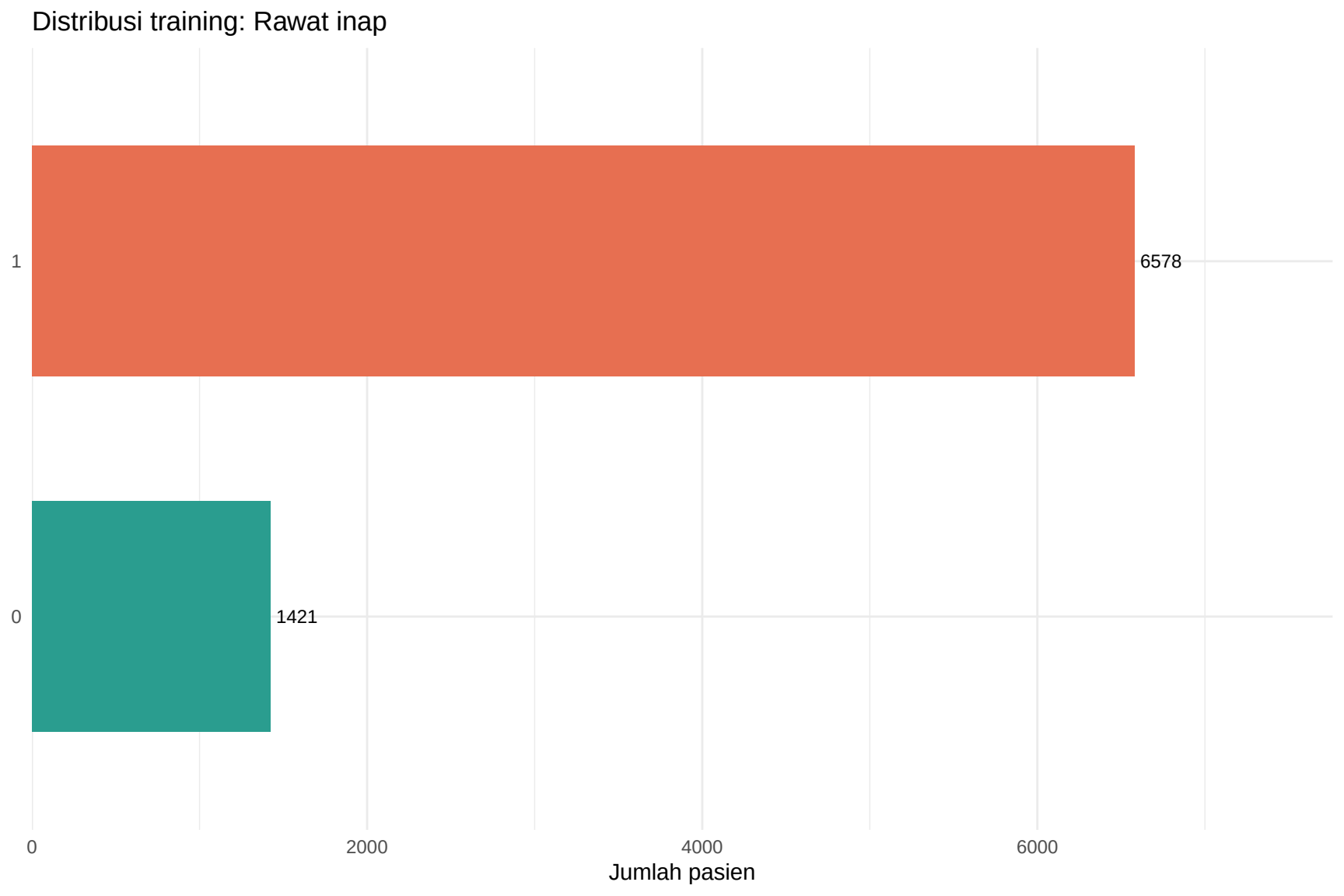
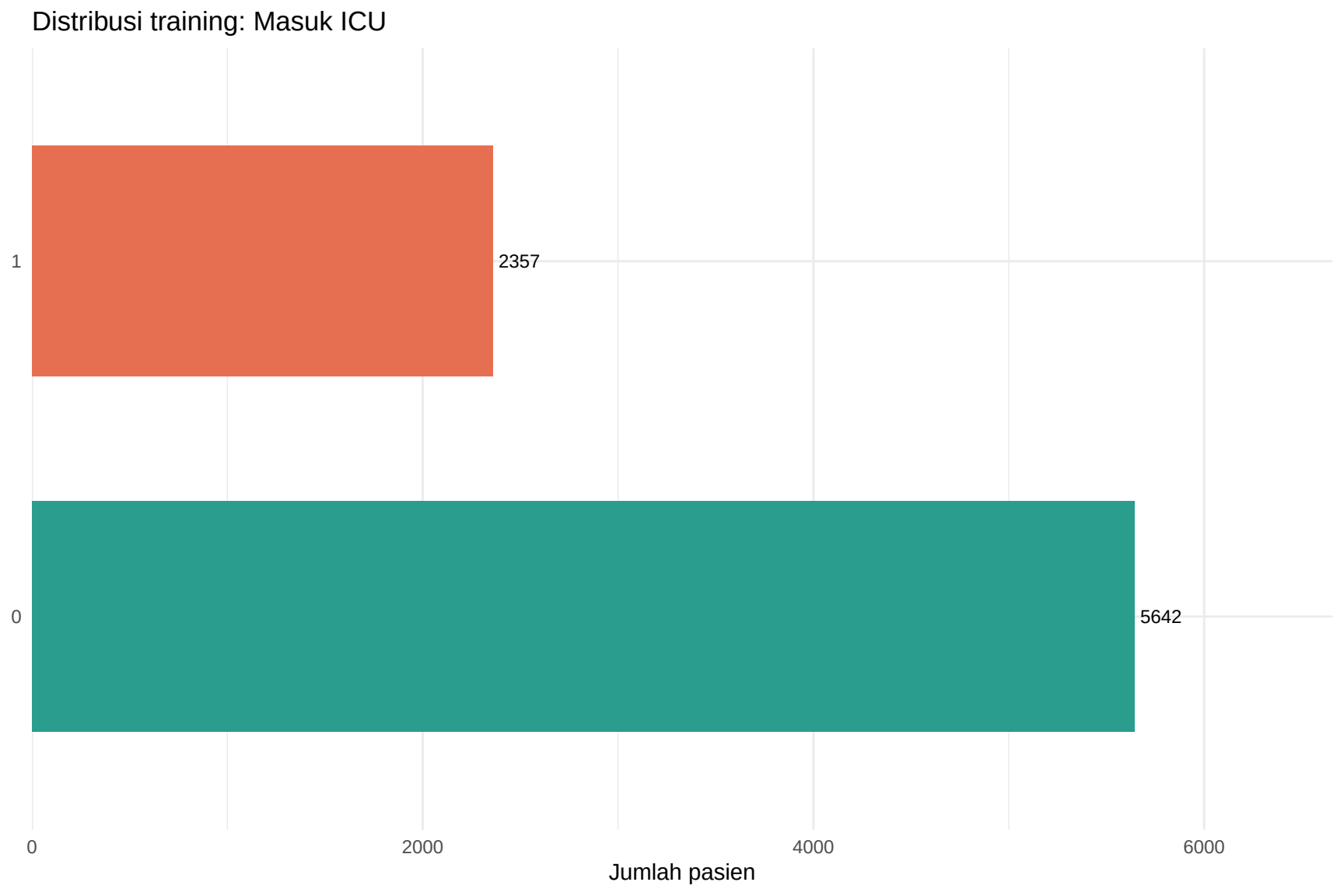

Distribusi training: Negara
Uruguay
USA
Sweden
South Korea
Slovenia
Russia
Paraguay
Panama
Kategori lain/jarang
Germany
Finland
Croatia
Colombia
China
Chile
Canada
Brazil
Bolivia
Argentina
255
287
653
692
621
669
245
302
10
620
655
655
278
624
289
280
282
307
275
0
200
400
600
800
Jumlah pasien

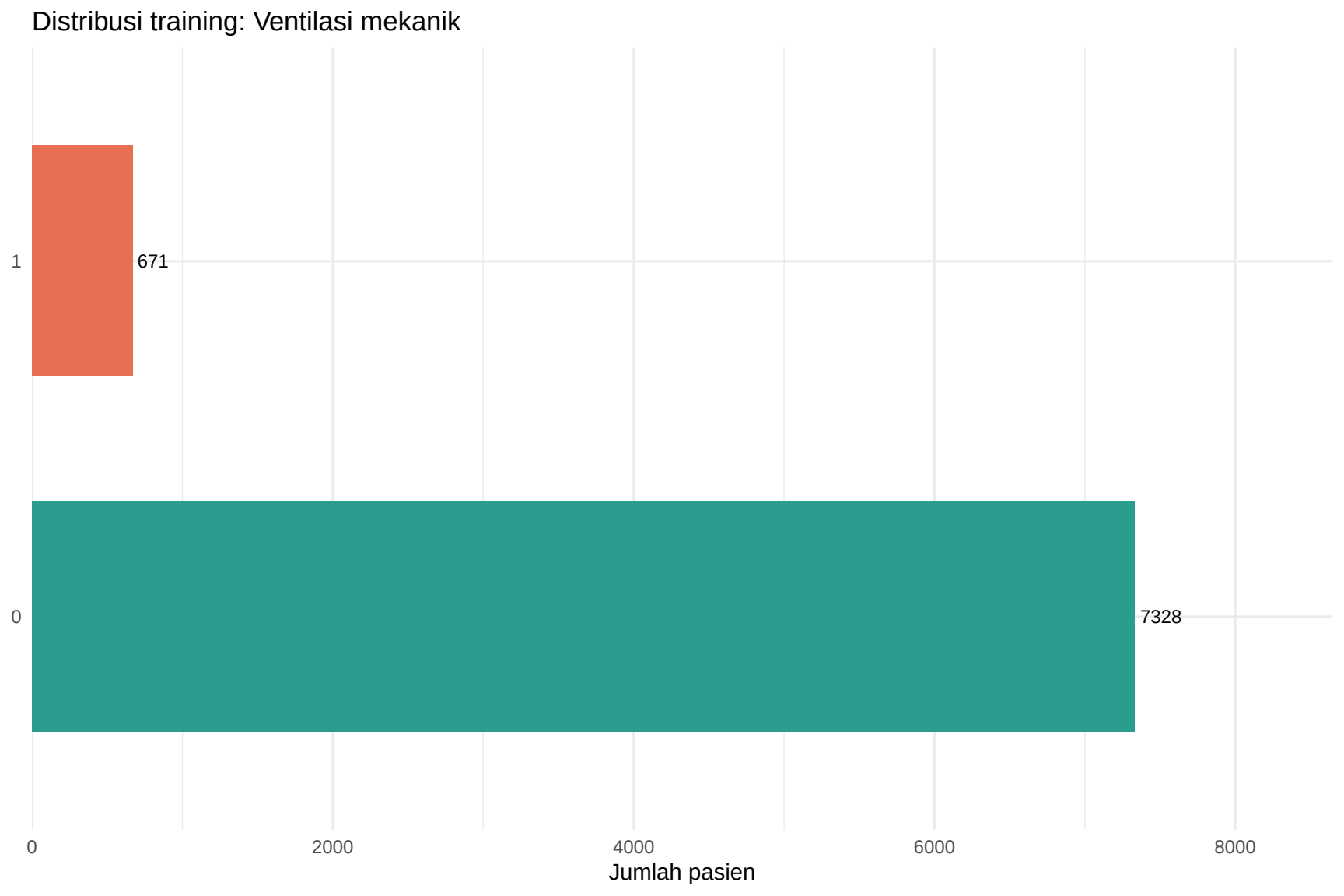

Distribusi training: Sindrom
HPS
HFRS
2810
5189
0
2000
4000
6000
Jumlah pasien

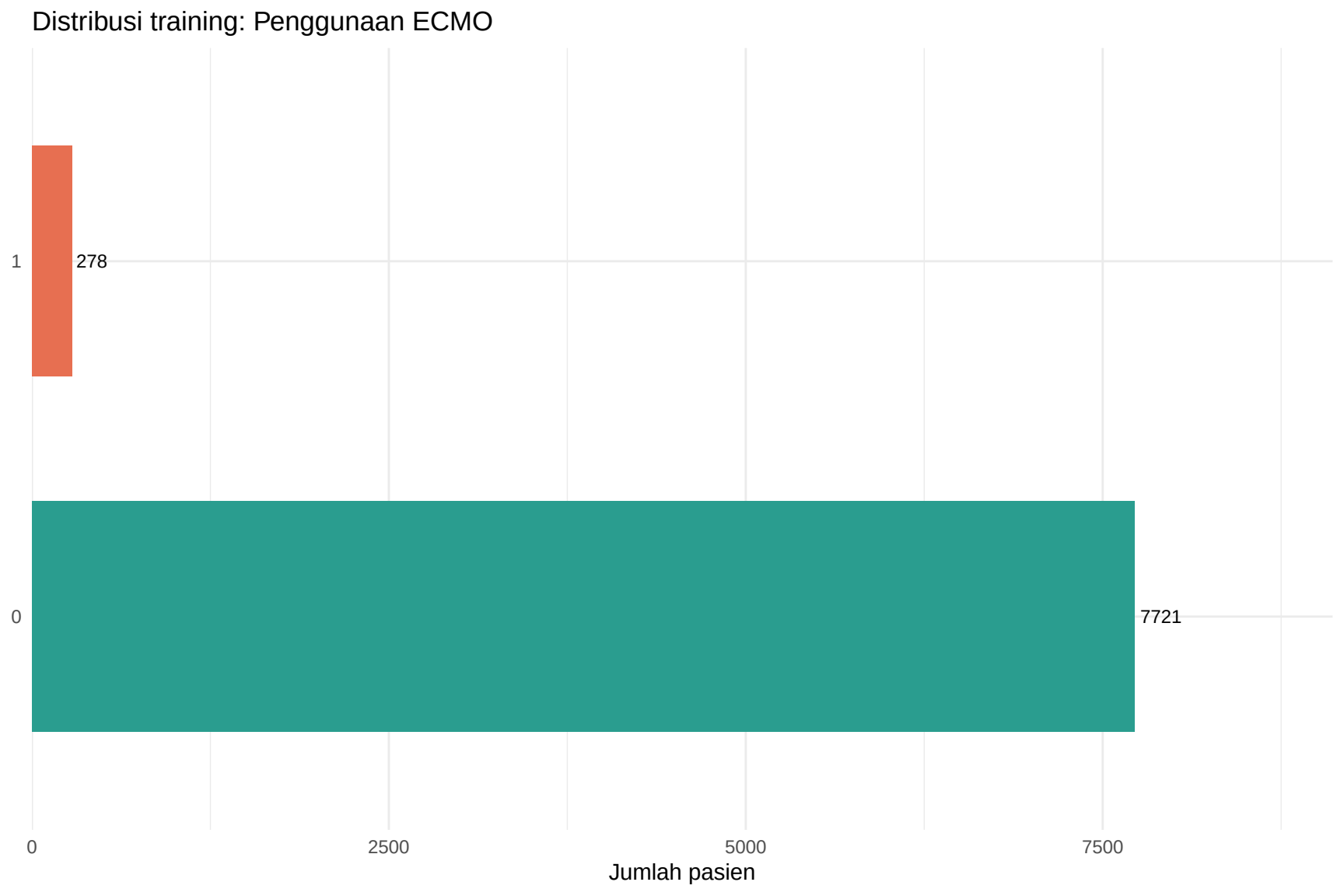

Distribusi training: Jenis kelamin
Male
Female
4903
3096
0
1000
2000
3000
4000
5000
Jumlah pasien

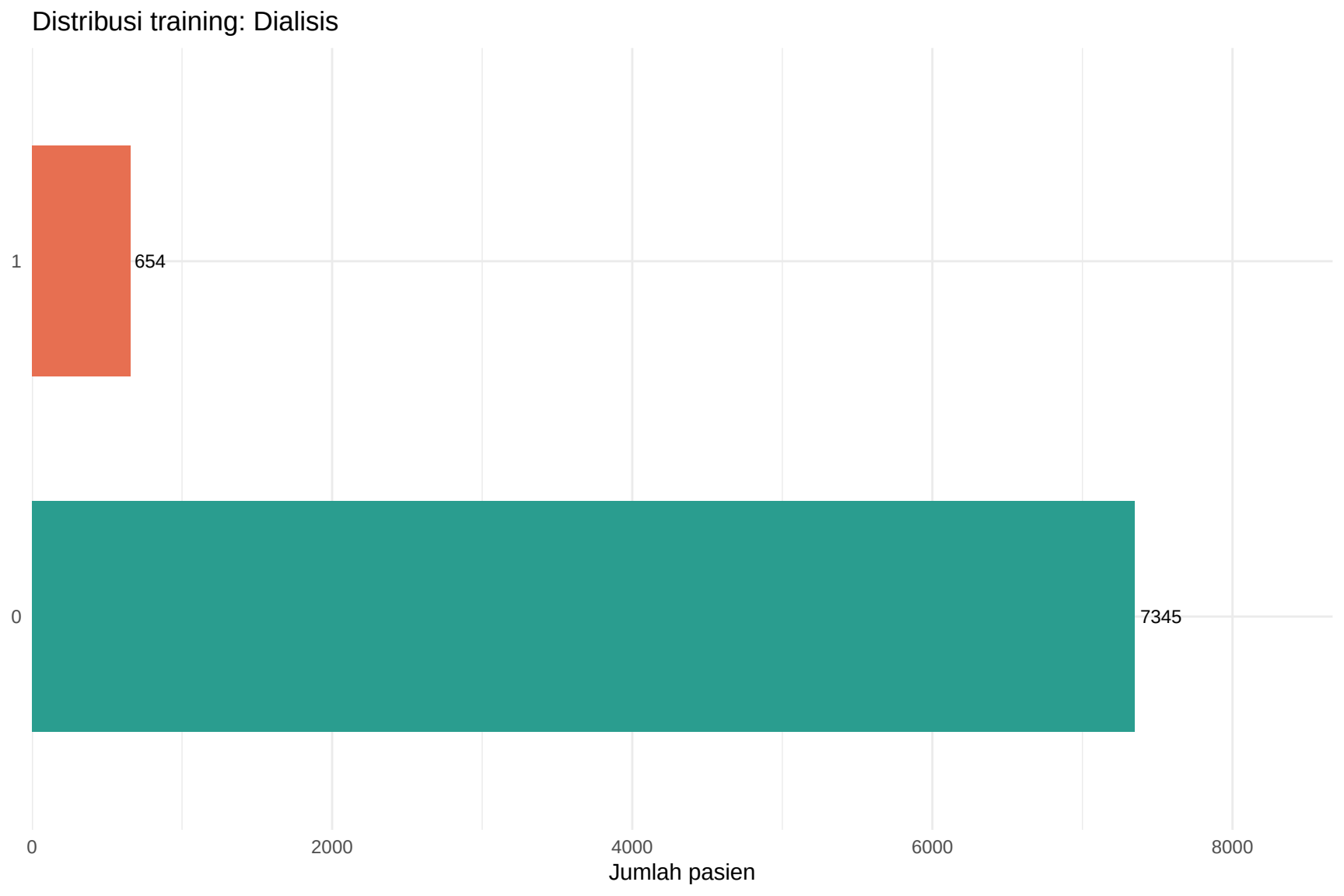

Distribusi training: Tingkat keparahan
Severe
Moderate
Mild
Critical
2286
2592
2018
1103
0
1000
2000
3000
Jumlah pasien

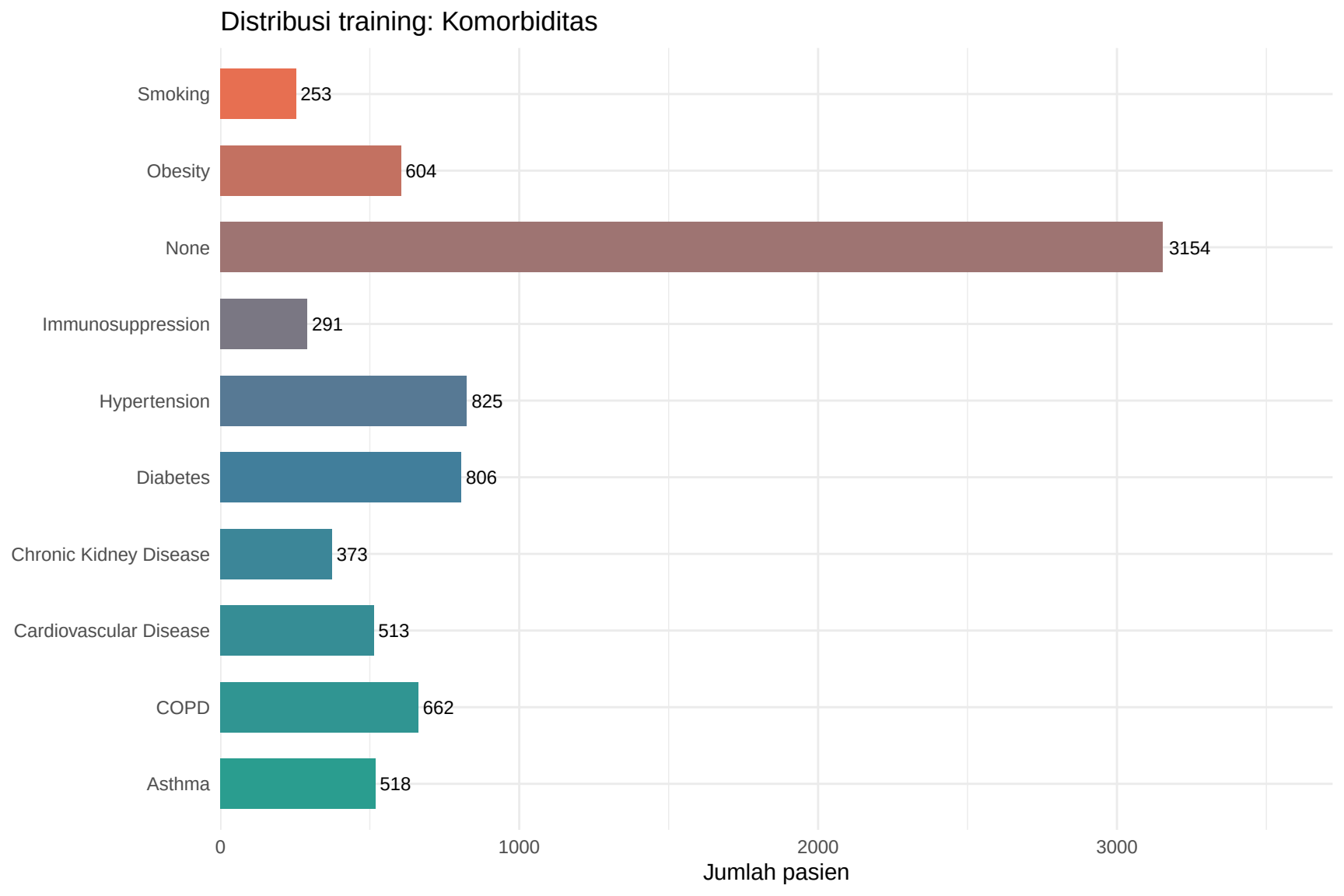

Distribusi training: Rawat inap
1
6578
0
1421
0
2000
4000
6000
Jumlah pasien

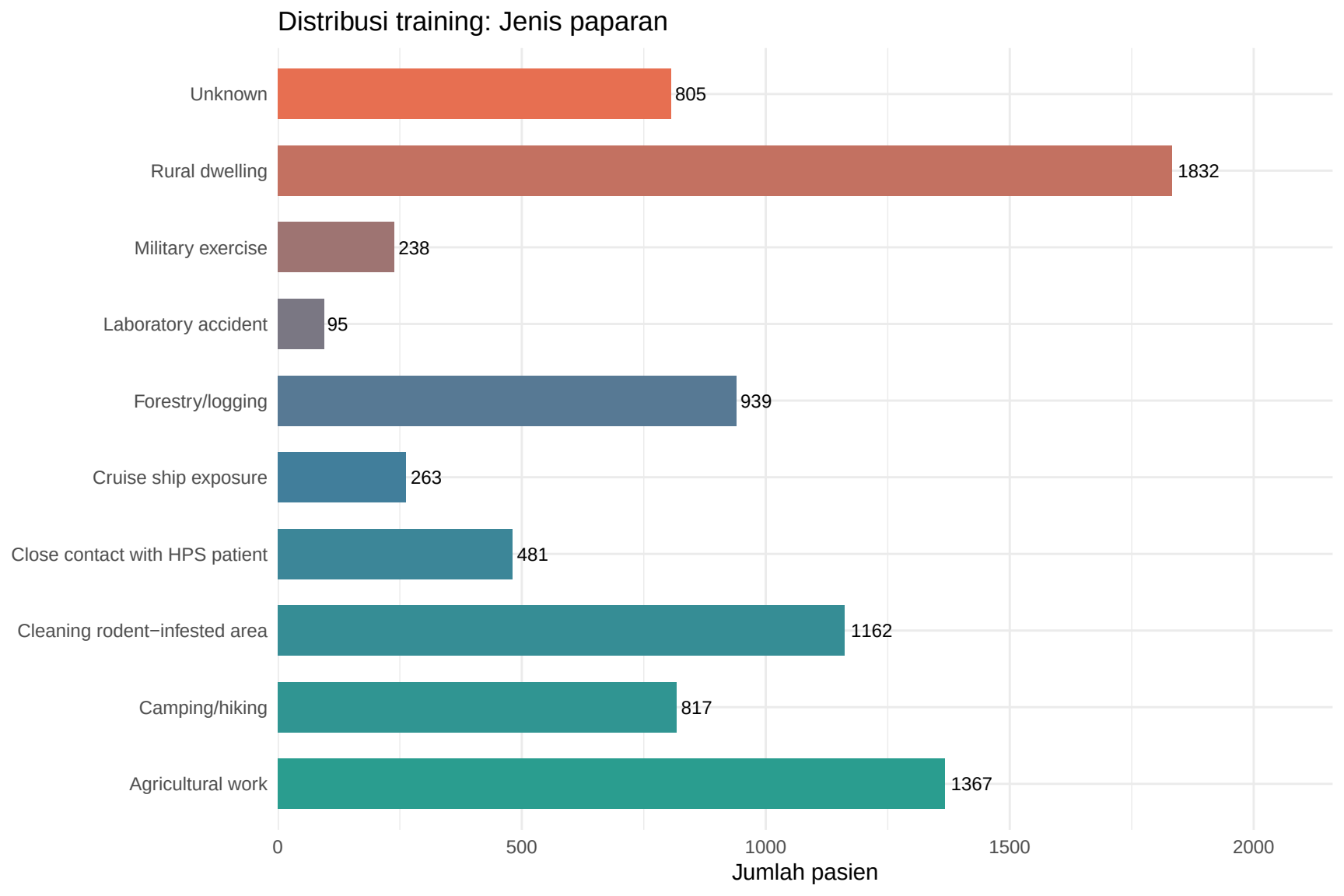

Distribusi training: Masuk ICU
1
0
2357
5642
0
2000
4000
6000
Jumlah pasien

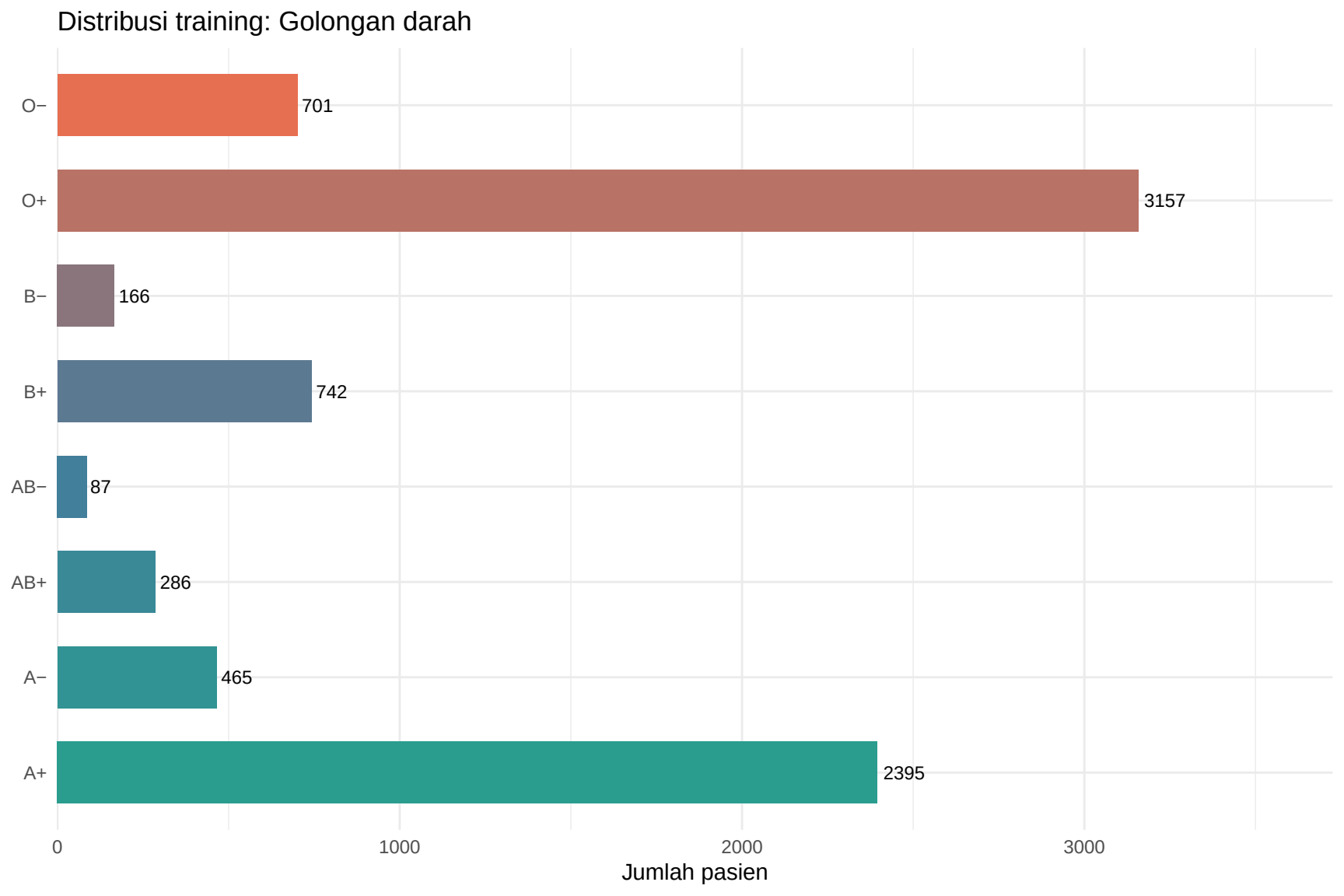

Distribusi training: Ventilasi mekanik
1
671
0
7328
0
2000
4000
6000
8000
Jumlah pasien

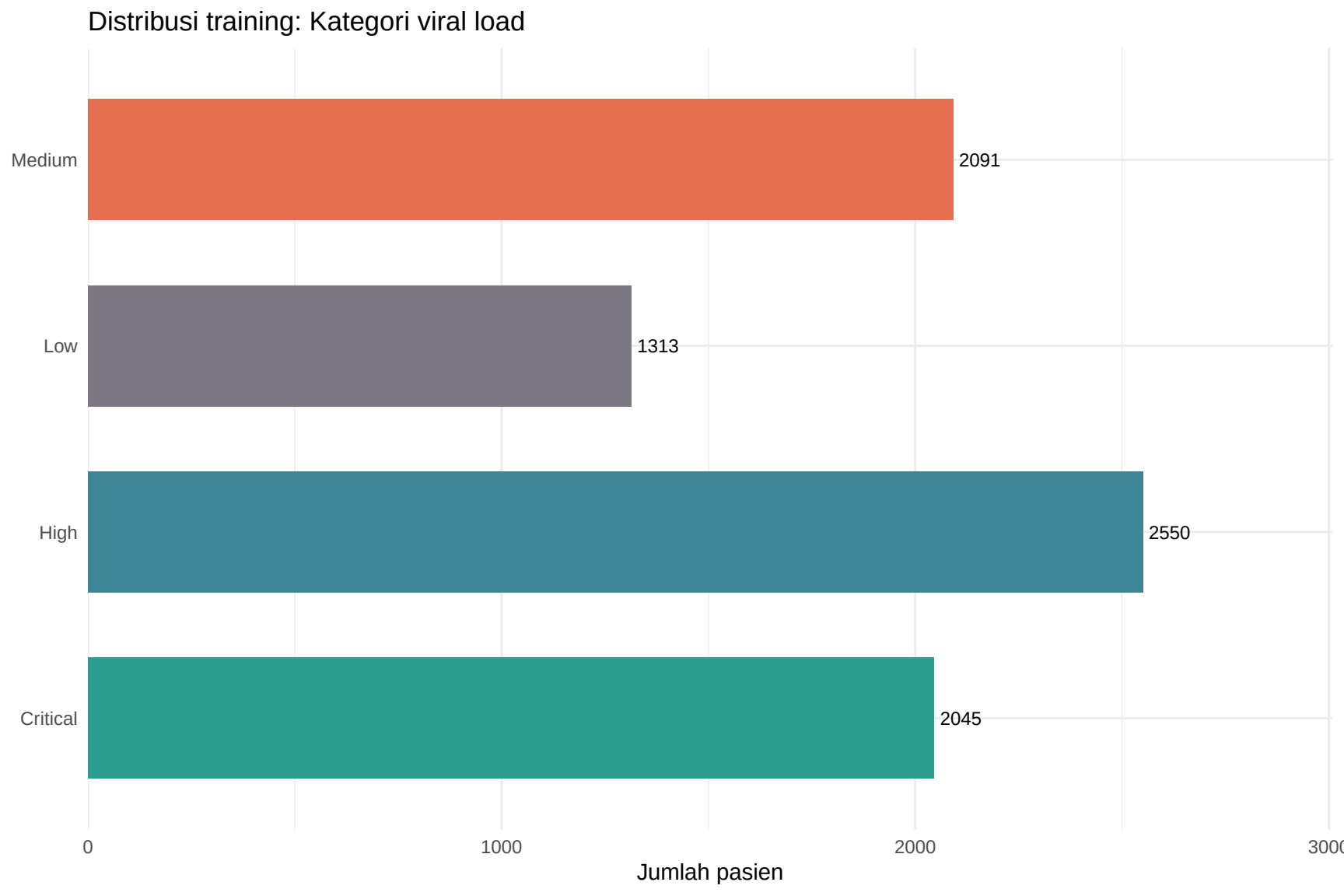

Distribusi training: Penggunaan ECMO
1
278
0
7721
0
2500
5000
7500
Jumlah pasien

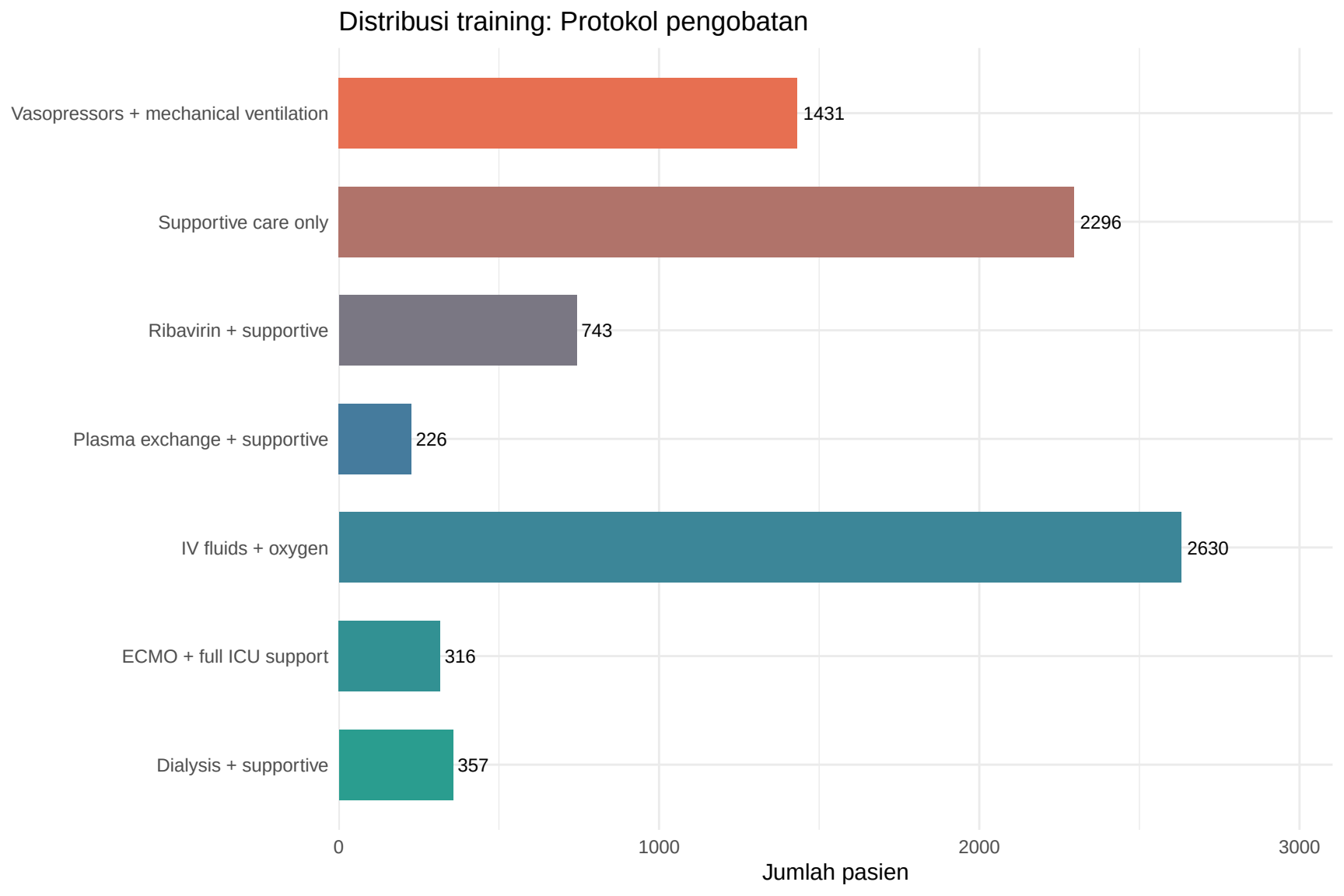

Distribusi training: Dialisis
1
654
0
7345
0
2000
4000
6000
8000
Jumlah pasien


Distribusi training: Komorbiditas
Smoking
253
Obesity
604
None
3154
Immunosuppression
291
Hypertension
825
Diabetes
806
Chronic Kidney Disease
373
Cardiovascular Disease
513
COPD
662
Asthma
518
0
1000
2000
3000
Jumlah pasien

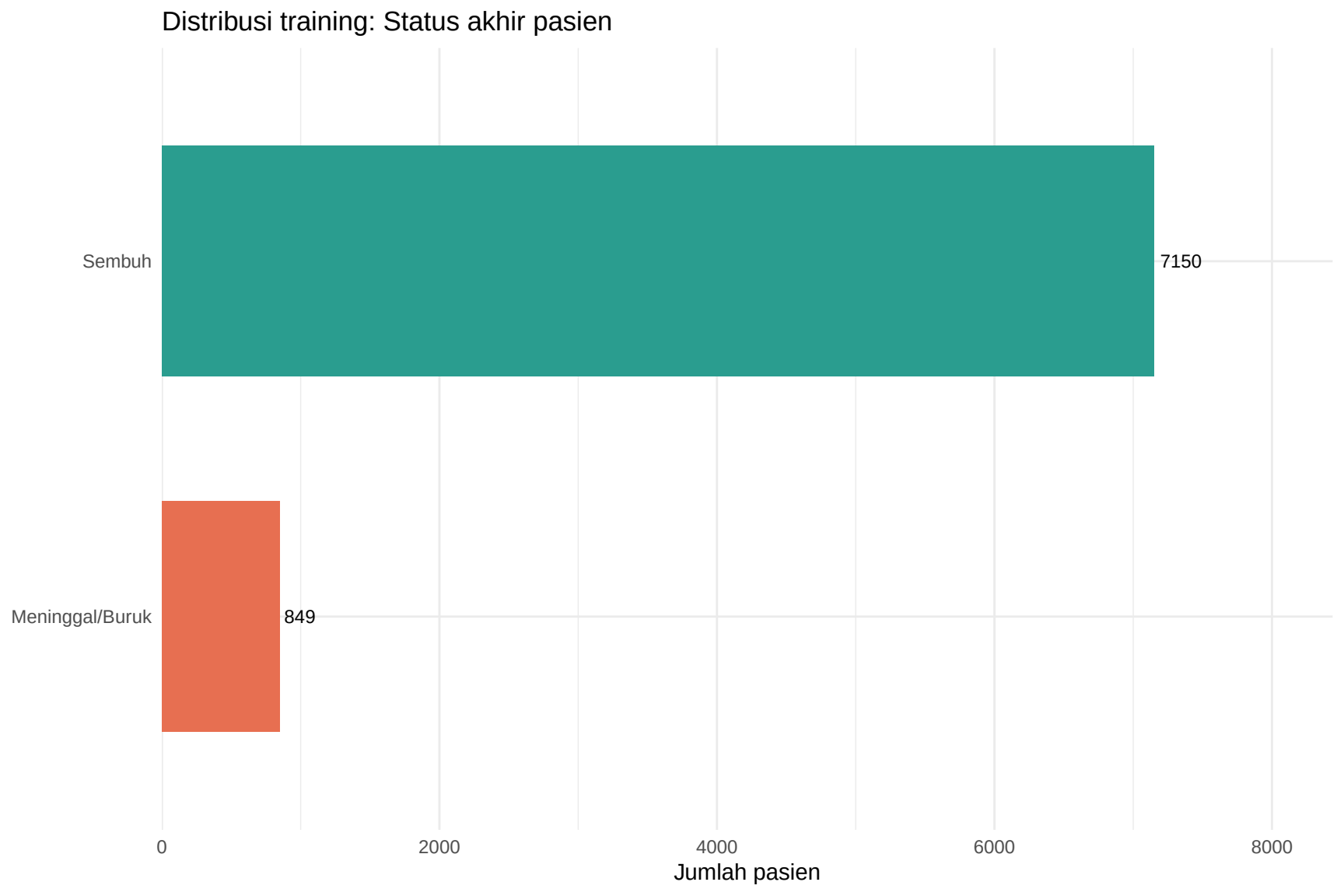

Distribusi training: Jenis paparan
Unknown
Rural dwelling
Military exercise
Laboratory accident
Forestry/logging
Cruise ship exposure
Close contact with HPS patient
Cleaning rodent−infested area
Camping/hiking
Agricultural work
805
1832
238
95
939
263
481
1162
817
1367
0
500
1000
1500
2000
Jumlah pasien

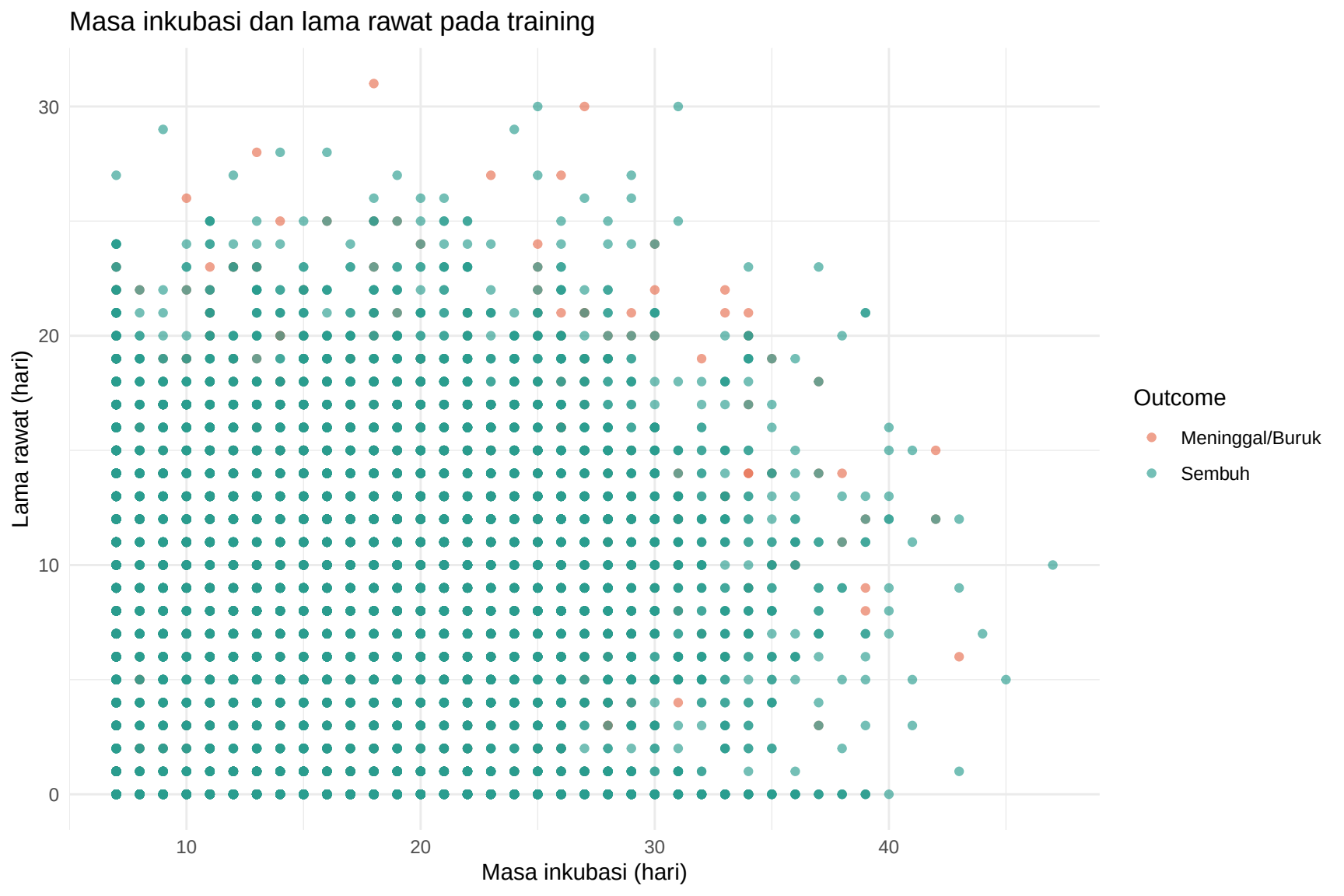

Distribusi training: Golongan darah
O−
O+
B−
B+
AB−
AB+
A−
A+
701
3157
166
742
87
286
465
2395
0
1000
2000
3000
Jumlah pasien


Distribusi training: Kategori viral load
Medium
Low
High
Critical
2091
1313
2550
2045
0
1000
2000
3000
Jumlah pasien

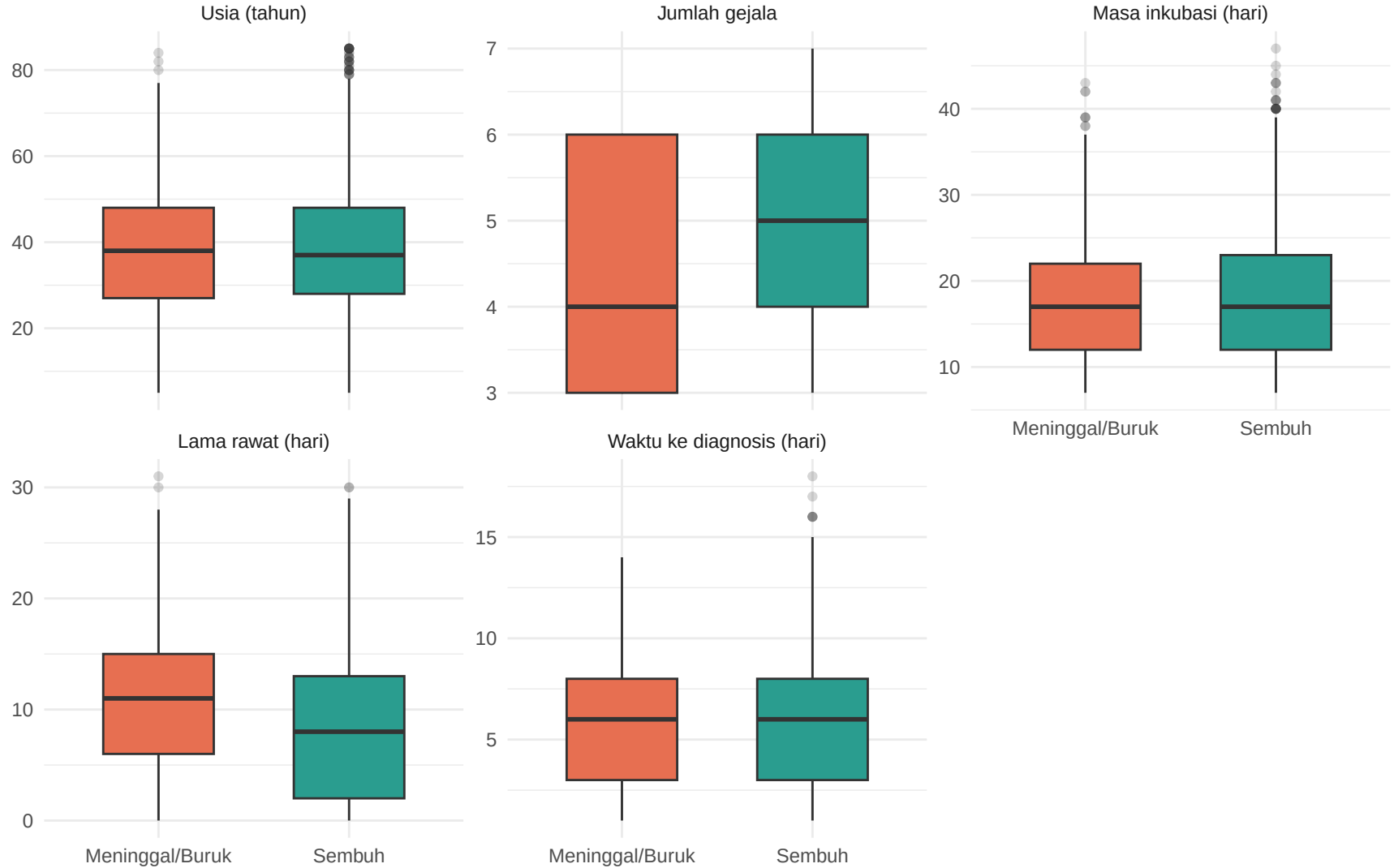

Distribusi training: Protokol pengobatan
Vasopressors + mechanical ventilation
1431
Supportive care only
2296
Ribavirin + supportive
743
Plasma exchange + supportive
226
IV fluids + oxygen
2630
ECMO + full ICU support
316
Dialysis + supportive
357
0
1000
2000
3000
Jumlah pasien

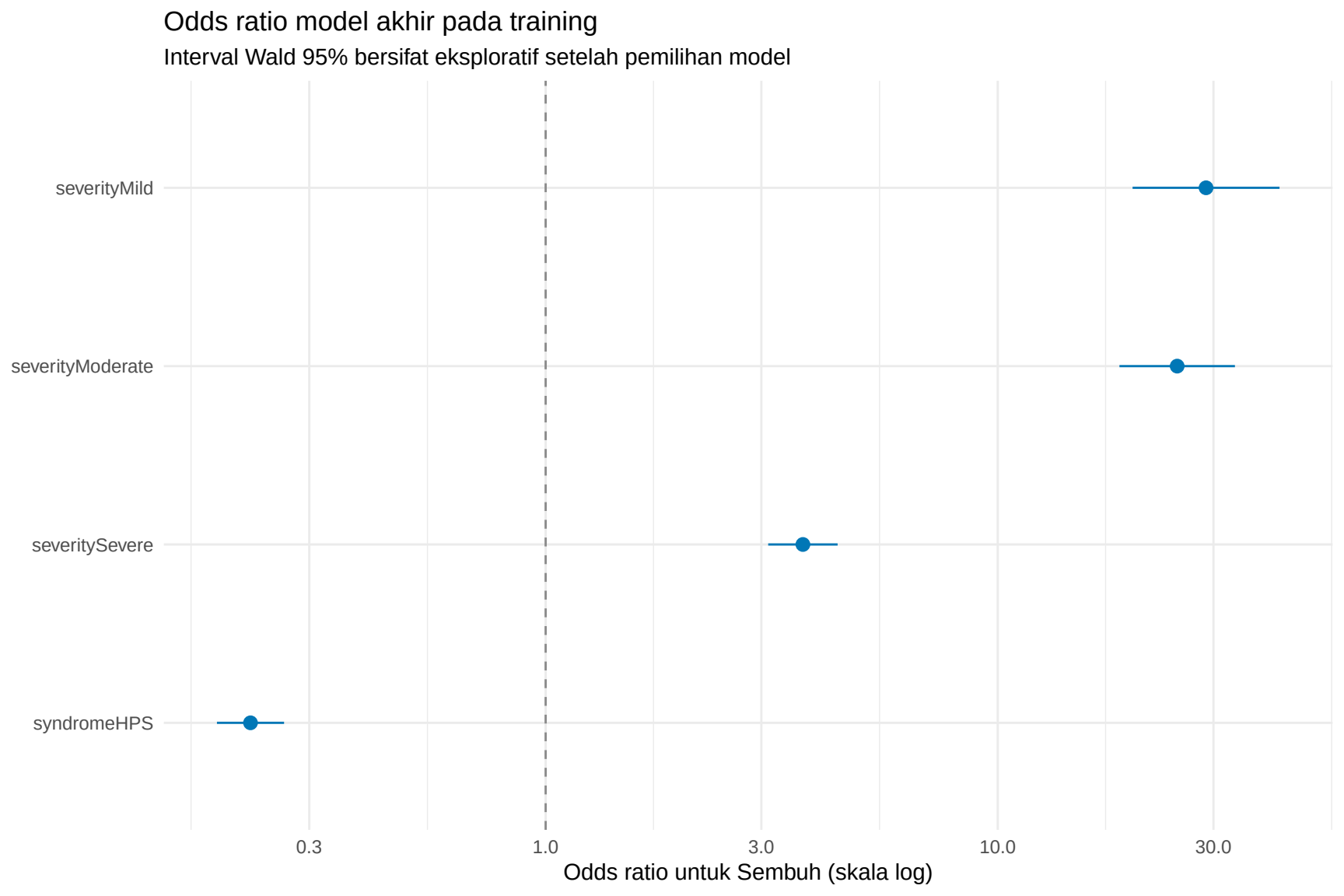

Distribusi training: Lokasi geografis
Urban
Rural
Remote
Peri−urban
835
3961
1979
1224
0
1000
2000
3000
4000
Jumlah pasien

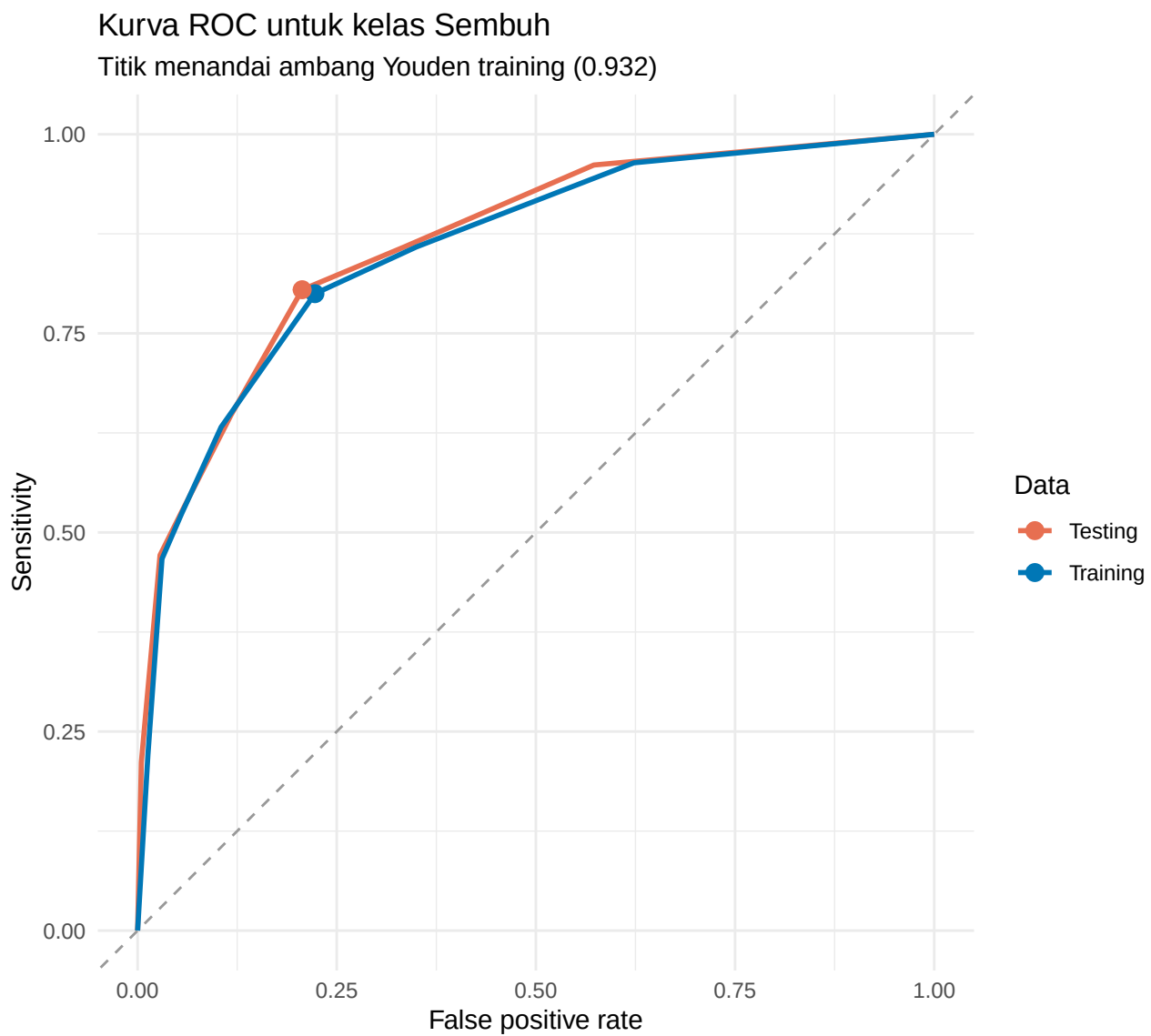

Distribusi training: Status akhir pasien
Sembuh
Meninggal/Buruk
7150
849
0
2000
4000
6000
8000
Jumlah pasien

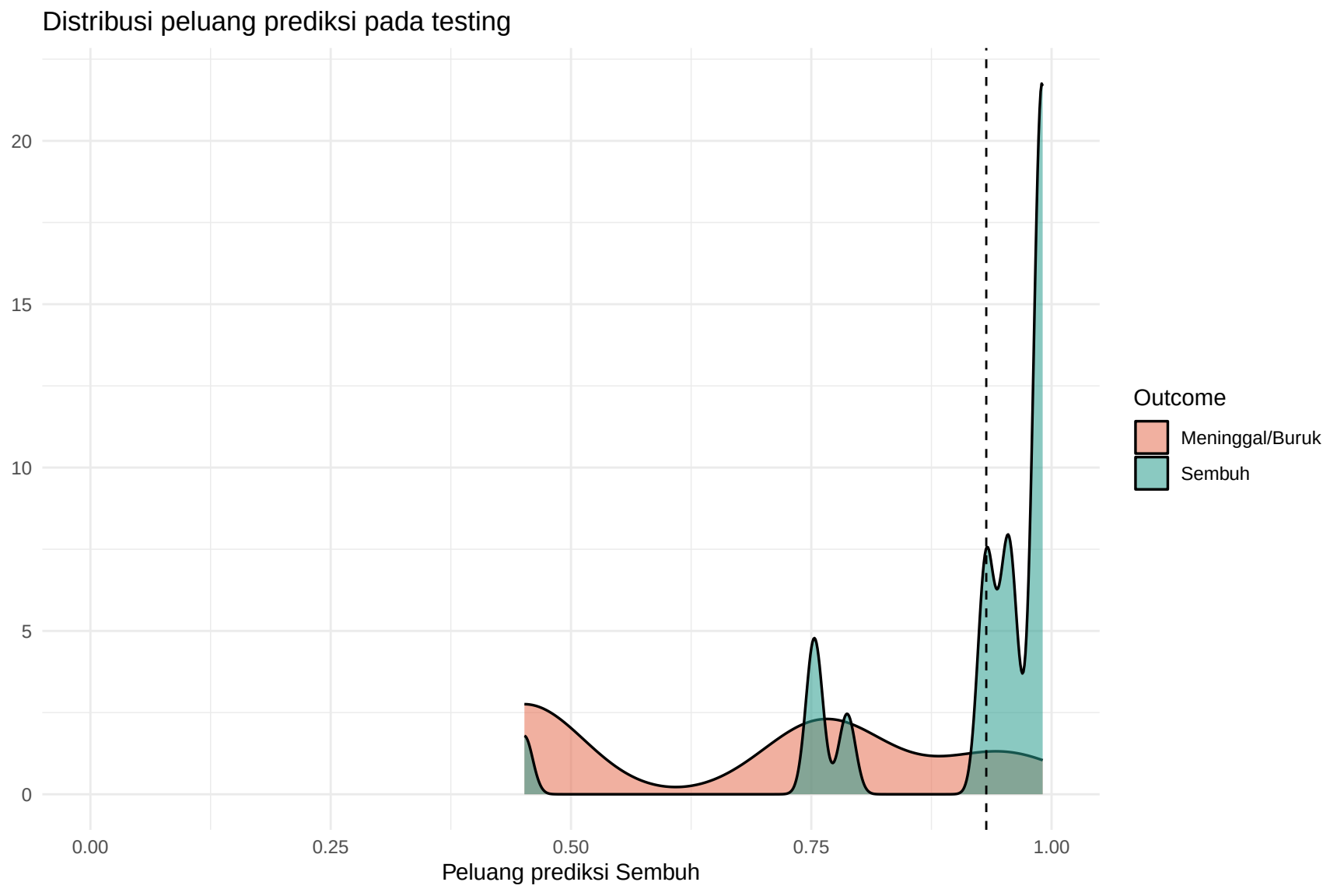

Masa inkubasi dan lama rawat pada training
Lama rawat (hari)
0
10
20
30
10
20
30
40
Masa inkubasi (hari)
Outcome
Meninggal/Buruk
Sembuh

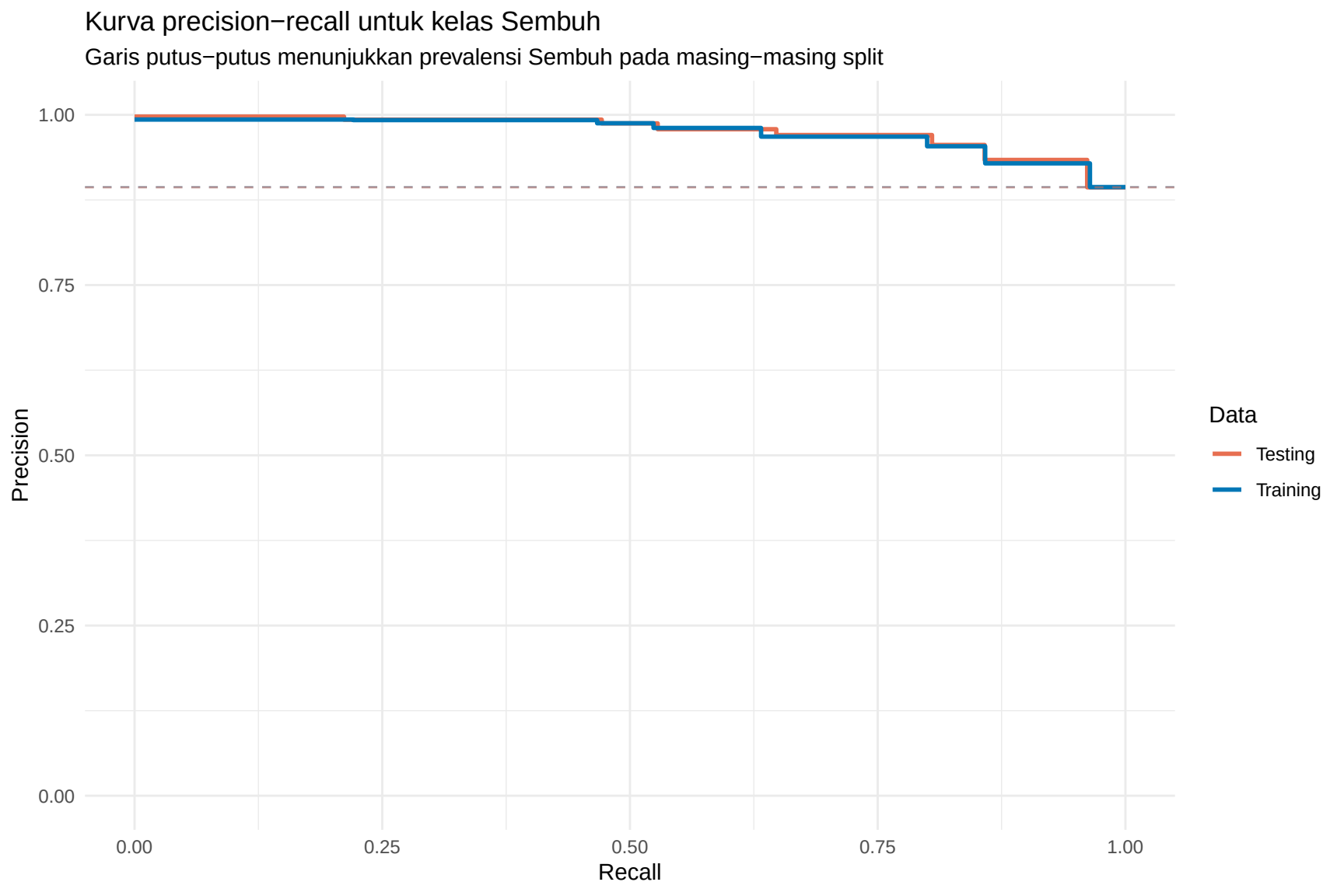
Sebaran variabel numerik menurut outcome pada training

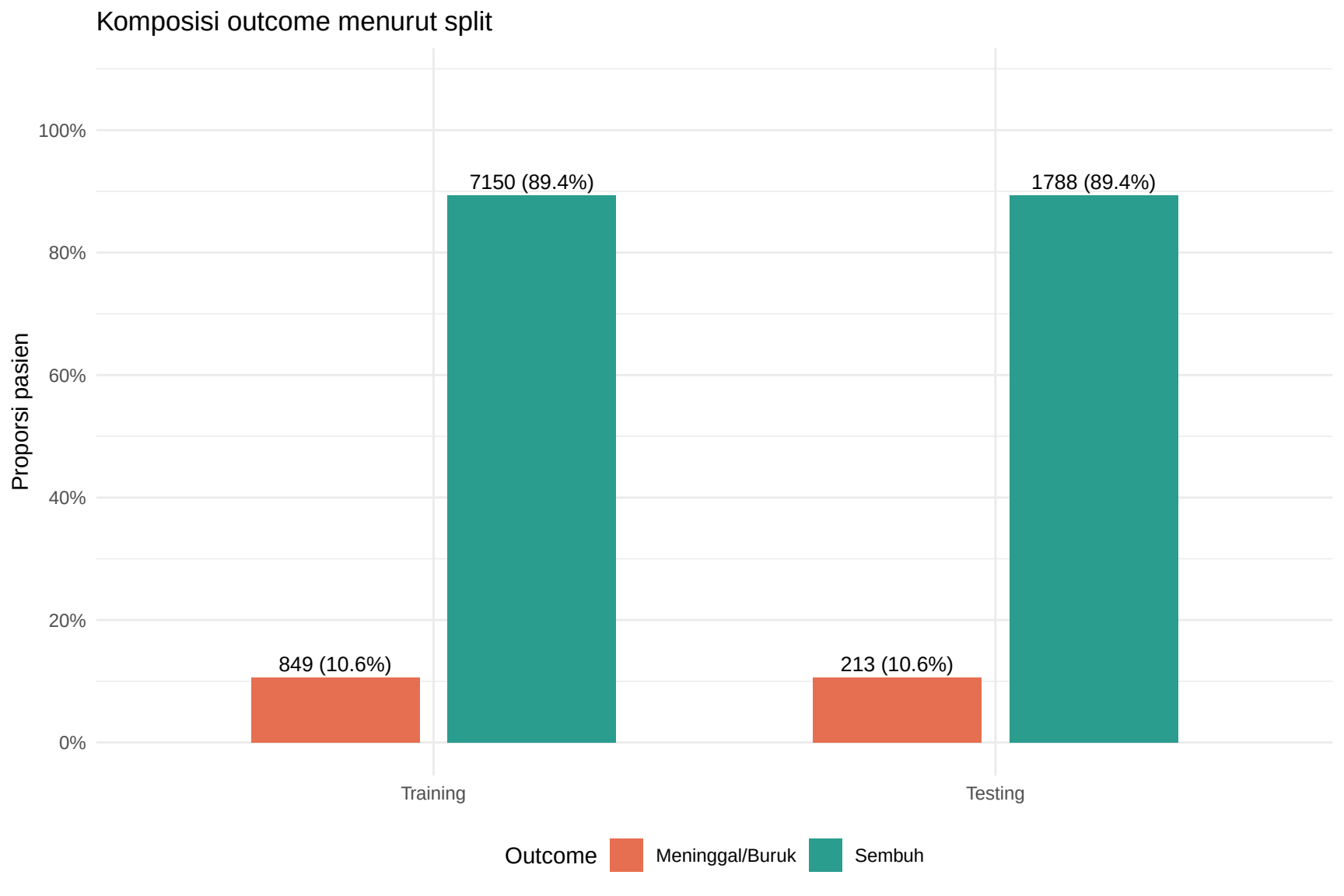


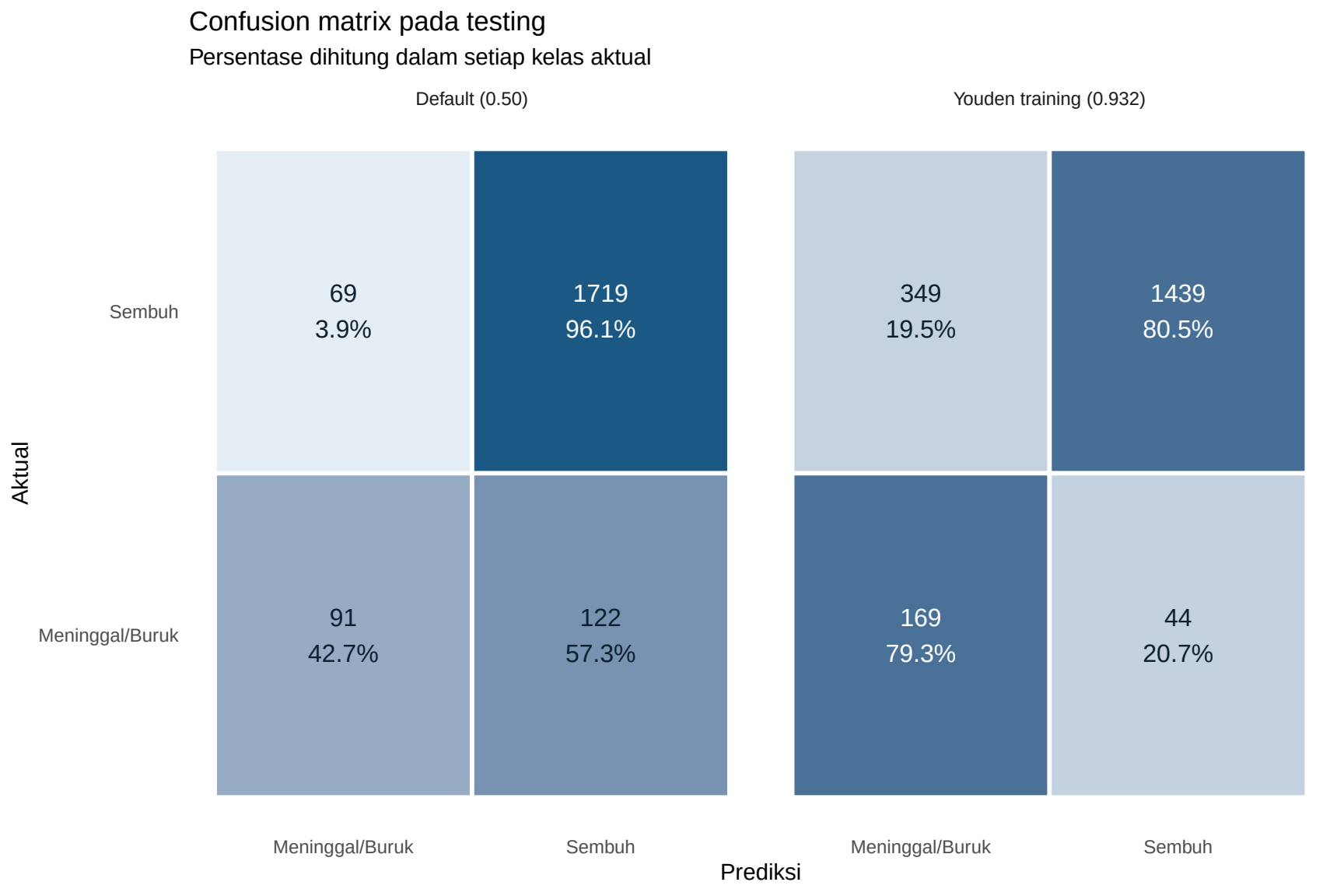

Odds ratio model akhir pada training
Interval Wald 95% bersifat eksploratif setelah pemilihan model
severityMild
severityModerate
severitySevere
syndromeHPS
0.3
1.0
3.0
10.0
30.0
Odds ratio untuk Sembuh (skala log)

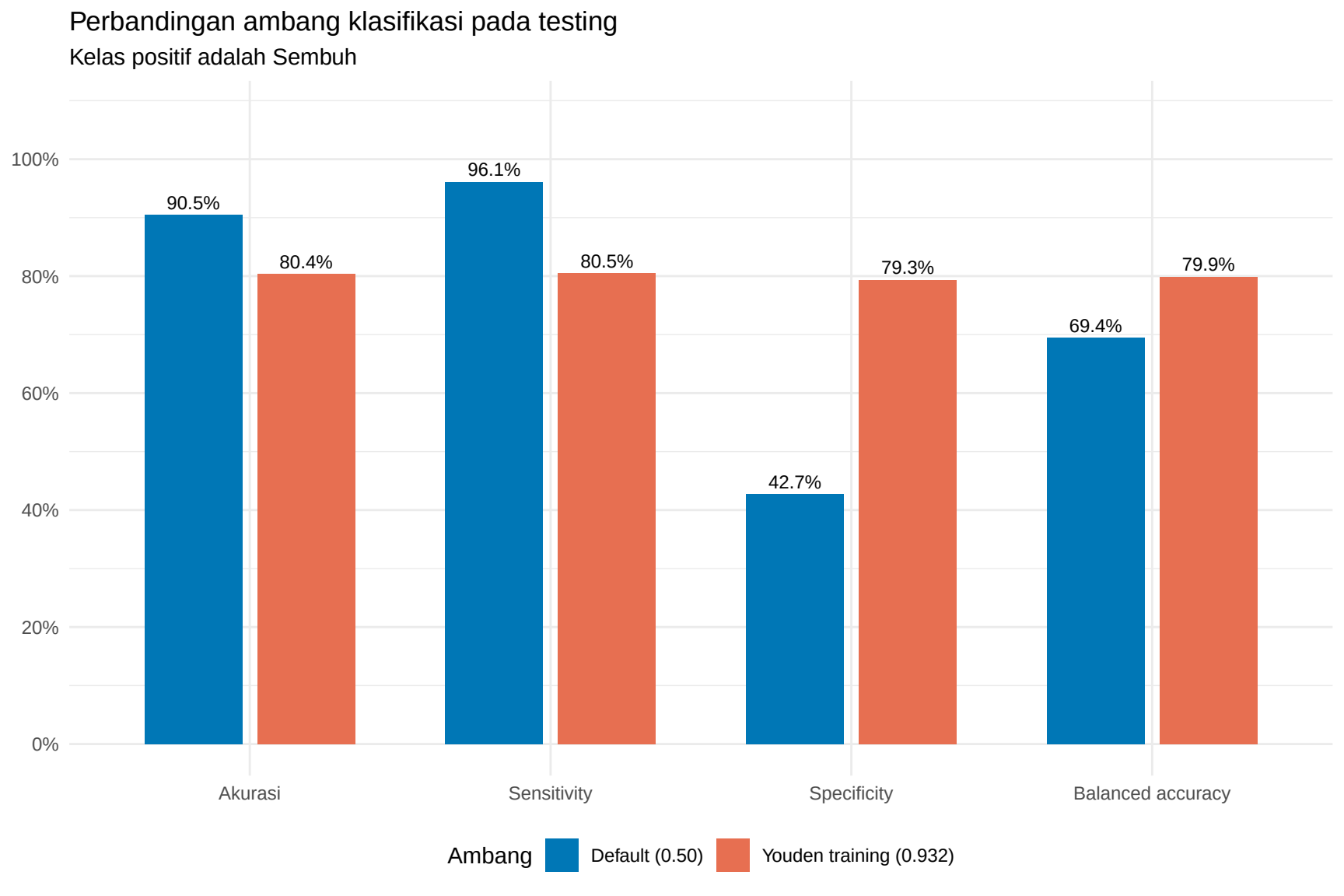

Kurva ROC untuk kelas Sembuh
Titik menandai ambang Youden training (0.932)
1.00
0.75
0.50
0.25
0.00
Sensitivity
0.00
0.25
0.50
0.75
1.00
False positive rate
Data
Testing
Training

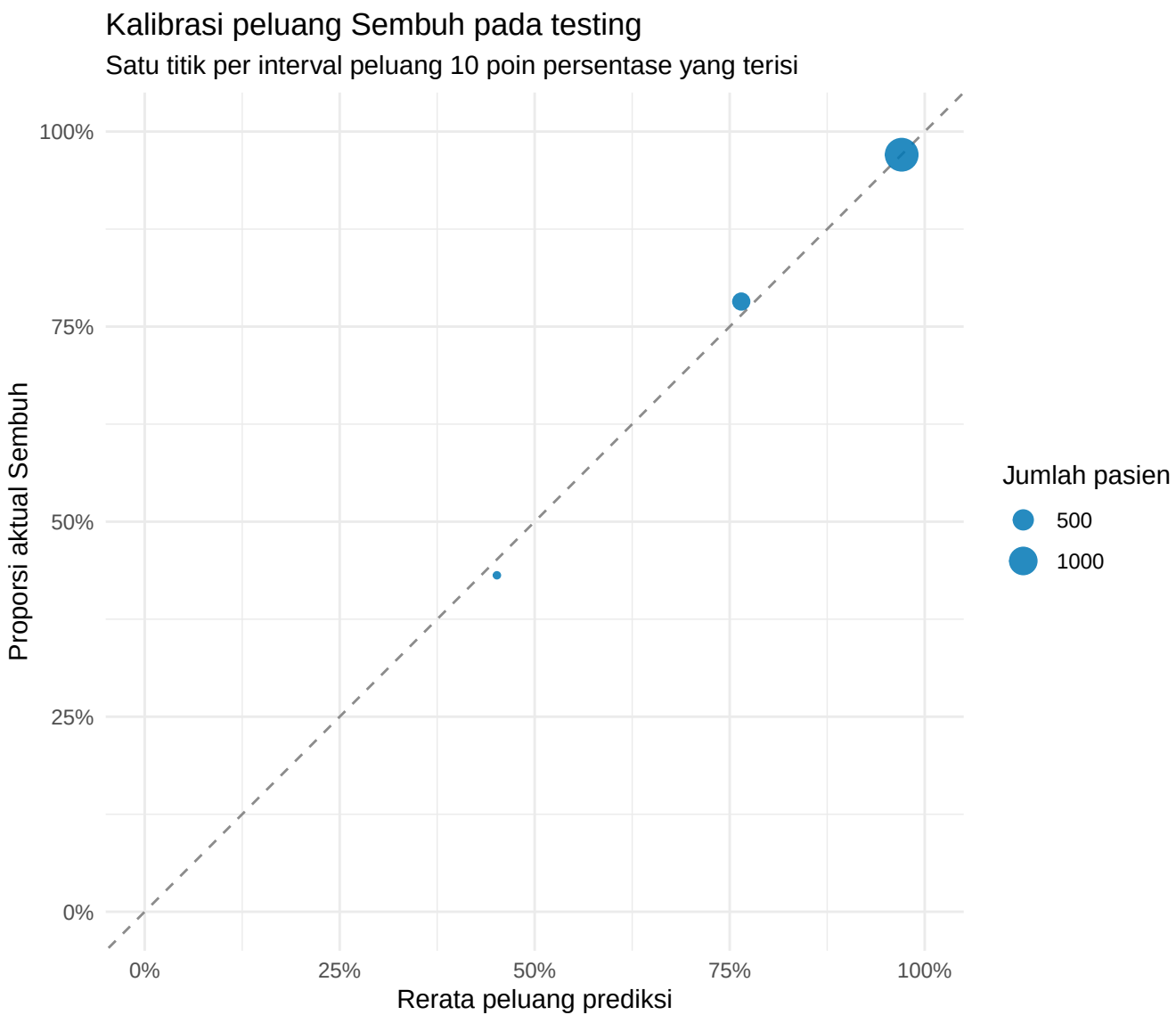

Distribusi peluang prediksi pada testing
20
15
10
5
0
0.00
0.25
0.50
0.75
1.00
Peluang prediksi Sembuh
Outcome
Meninggal/Buruk
Sembuh


Kurva precision−recall untuk kelas Sembuh
Garis putus−putus menunjukkan prevalensi Sembuh pada masing−masing split
Precision
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Recall
Data
Testing
Training


Komposisi outcome menurut split
Proporsi pasien
100%
80%
60%
40%
20%
0%
7150 (89.4%)
1788 (89.4%)
849 (10.6%)
213 (10.6%)
Training
Testing
Outcome
Meninggal/Buruk
Sembuh


Confusion matrix pada testing
Persentase dihitung dalam setiap kelas aktual
Default (0.50)
Youden training (0.932)
Sembuh
Meninggal/Buruk
Aktual
69
3.9%
1719
96.1%
91
42.7%
122
57.3%
349
19.5%
1439
80.5%
169
79.3%
44
20.7%
Meninggal/Buruk
Sembuh
Meninggal/Buruk
Sembuh
Prediksi


Perbandingan ambang klasifikasi pada testing
Kelas positif adalah Sembuh
100%
80%
60%
40%
20%
0%
90.5%
80.4%
96.1%
80.5%
42.7%
79.3%
69.4%
79.9%
Akurasi
Sensitivity
Specificity
Balanced accuracy
Ambang
Default (0.50)
Youden training (0.932)


Kalibrasi peluang Sembuh pada testing
Satu titik per interval peluang 10 poin persentase yang terisi
Proporsi aktual Sembuh
100%
75%
50%
25%
0%
0%
25%
50%
75%
100%
Rerata peluang prediksi
Jumlah pasien
500
1000